

PWM-LDCT v0.5: LIDC-only first release

Mayo Clinic EPD release (release scope (removed 2026-08-21))

LIDC-IDRI Source (public, CC BY 3.0)

Full-dose DICOM scans, $n = 1,010$ patients (4 vendors: GE / Siemens / Philips / Toshiba)
Four radiologist module annotations (XML), 3,643 files
Sole v0.5 source; no new acquisitions

v1.0 Roadmap (validation subset)

- AAPM 2016 Low-Dose CT Grand Challenge
- 10 training patients, Siemens, 1 mm B30
- Real paired full-dose + quarter-dose (Mayo noise insertion)
- Same harmonization schema
- Simulation-vs-reference discrepancy measurable

Preprocessing Pipelines (Docker, deterministic)

HU calibration | pixel-spacing normalization | thin-slice selection
Reconstruction-kernel preservation | DICOM de-identification re-verification
Low-dose simulation: Poisson photon noise + electronic floor,
dose ratios $r \in \{0.10, 0.25, 0.50\}$ (projection-domain forward model)

Unified Python Loader (pip install pwm_ldct_loader)

Single schema across sources | deterministic annotations, metadata-pinned

Harmonized, Content-Addressed Records

HDF5 shards (regenerated within per-voxel tolerance)
Multi-task annotations: majority-vote + raw per-reader
Simulated low-dose images at $r \in \{0.10, 0.25, 0.50\}$
Harmonized metadata | deterministic patient-level splits
Evaluation: PSNR / SSIM / LPIPS + task-based detectability (BandER)

SHA-256 content manifest (version-pinned) → PhysioNet DOI (primary) + Zenodo mirror + GitHub release

build →

load →

(内容由AI生成，仅供参考)